

# Math 299 Exercises

Lance Remigio

October 8, 2025

## Contents

|          |                                 |          |
|----------|---------------------------------|----------|
| <b>1</b> | <b>Week 1</b>                   | <b>1</b> |
| 1.1      | Zorn's Lemma . . . . .          | 1        |
| 1.2      | Hahn-Banach Theorem . . . . .   | 1        |
| <b>2</b> | <b>Week 7: Reflexive Spaces</b> | <b>3</b> |

## 1 Week 1

### 1.1 Zorn's Lemma

**Problem 1** (4.1.2). Let  $X$  be the set of all real-valued functions on  $X$  on the interval  $[0, 1]$ , and let  $x \leq y$  to mean that  $x(t) \leq y(t)$  for all  $t \in [0, 1]$ . Show that this defines a partial ordering. Is it total ordering? Does  $X$  have a maximal elements?

**Proof.** Define  $F = \{\text{all real-valued functions on the interval } [0, 1]\}$ . First, we show reflexivity. Let  $x \in F$ . Then for all  $t \in [0, 1]$ ,  $x(t) \leq x(t)$ . Hence,  $x \leq x$  and so  $F$  contains the reflexive property. For antisymmetry, suppose  $x \leq y$  and  $y \leq x$ . Then for all  $t \in [0, 1]$ , we have  $x(t) \leq y(t)$  and  $y(t) \leq x(t)$ . Then  $x(t) = y(t)$  (using the partial ordering of the real numbers) for all  $t \in [0, 1]$ . Then  $x = y$  and so antisymmetry is satisfied. For transitivity, suppose  $x \leq y$  and  $y \leq z$ . Then  $x(t) \leq y(t)$  and  $y(t) \leq z(t)$  for all  $t \in [0, 1]$ . Then  $x(t) \leq z(t)$  for all  $t \in [0, 1]$  using the transitivity of the real numbers. Hence,  $x \leq z$  and so transitivity is satisfied.

$F$  is totally ordered, but has no maximal elements. ■

**Problem 2** (4.1.5). Prove that a finite partially ordered set  $A$  has at least one maximal element.

**Proof.** ■

### 1.2 Hahn-Banach Theorem

**Problem 3** (Existence of a Sublinear Functional). Show that a sublinear functional  $p$  satisfies  $p(0) = 0$  and  $p(-x) \geq -p(x)$ .

**Proof.** Define  $P : X \rightarrow \mathbb{R}$  by

$$p(x) = \|x\|$$

where  $X$  is some vector space. Then for  $x = 0$ , we have

$$p(0) = \|0\| = 0.$$

Note that for all  $x \in X$ , we have

$$p(-x) = \|-x\| = |-1|\|x\| = 1 \cdot \|x\| = \|x\| > 0$$

and  $p(-x) = p(x)$ . Now, note that for all  $x \in X$

$$-\|x\| \leq \|x\| \iff -p(x) \leq p(x) = p(-x).$$

Thus, for all  $x \in X$ ,  $p(-x) \geq -p(x)$ . ■

**Problem 4 (Convex Set).** If  $p$  is a sublinear functional on a vector space  $X$ , show that  $M = \{x : p(x) \leq \gamma, \gamma > 0 \text{ fixed}\}$ , is a convex set.

**Proof.** Assume that  $p$  is a sublinear functional on a vector space  $X$ . Let  $W = \{v = \alpha y + (1 - \alpha)z : 0 \leq \alpha \leq 1\}$ . Our goal is to show that  $W \subseteq M$ . To this end, let  $v \in W$  be arbitrary. Then for some  $y, z \in M$ , we have  $v = \alpha y + (1 - \alpha)z$  where  $\alpha \in \mathbb{R}$ . Since  $y, z \in M$ , we have  $p(y) \leq \gamma$  and  $p(z) \leq \gamma$  where  $\gamma > 0$  is fixed. Our goal is to show that  $p(v) \leq \gamma$  for fixed  $\gamma$ . Using the sublinearity of  $p$ , we get that

$$\begin{aligned} p(v) &= p(\alpha y + (1 - \alpha)z) \\ &\leq p(\alpha y) + p((1 - \alpha)z) \\ &= |\alpha|p(y) + |(1 - \alpha)|p(z) \\ &= \alpha p(y) + (1 - \alpha)p(z) & (0 \leq \alpha \leq 1) \\ &\leq \alpha\gamma + (1 - \alpha)\gamma \\ &= \gamma. \end{aligned}$$

Hence, we conclude that  $W \subseteq M$  since  $v$  was an arbitrary element and so  $M$  is a convex set. ■

**Problem 5.** Let  $p$  be a sublinear functional on a real vector space  $X$ . Let  $f$  be defined on  $Z = \{x \in X : x = \alpha x_0, \alpha \in \mathbb{R}\}$  by  $f(x) = \alpha p(x_0)$  with fixed  $x_0 \in X$ . Show that  $f$  is a linear functional on  $Z$  satisfying  $f(x) \leq p(x)$ .

**Proof.** First, we will show that  $f$  is a linear functional on  $Z$ . Let  $u, v \in Z$ . Then  $u = \alpha_1 x_0$  and  $v = \alpha_2 x_0$  for  $\alpha_1, \alpha_2 \in \mathbb{R}$ . Let  $\delta \in \mathbb{R}$ . Observe that

$$f(x) = \alpha p(x_0) = p(\alpha x_0).$$

Using this observation, we have

$$\begin{aligned} \delta f(u) + f(v) &= \delta \alpha_1 p(x_0) + \alpha_2 p(x_0) \\ &= (\delta \alpha_1 + \alpha_2) p(x_0) \\ &= p(\delta \alpha_1 x_0 + \alpha_2 x_0) \\ &= p(\delta u + v) \\ &= f(\delta u + v). \end{aligned}$$

Hence,  $f$  is a linear functional. Using our observation again, we can also see that for  $\alpha > 0$ ,  $f(x) = p(x)$  and so  $f(x) \leq p(x)$  for all  $x \in Z$ . Clearly, the inequality holds if  $\alpha = 0$ . Now, suppose  $\alpha < 0$ . Then

$$f(x) = \alpha p(x_0) = |\alpha| p(x_0) = p(|\alpha| x_0) = p(x)$$

and so  $f(x) \leq p(x)$  for all  $x \in Z$ . ■

**Problem 6.** If  $p$  is a sublinear on a real vector space  $X$ , show that there exists a linear functional  $\tilde{f}$  on  $X$  such that

$$-p(-x) \leq \tilde{f}(x) \leq p(x).$$

**Proof.** Define  $Z$  as in the set in the previous problem and define  $f(x) = \alpha p(x_0)$  with fixed  $x_0 \in X$ . Using the same problem, we proved that  $f$  defines linear functional such that  $f(x) \leq p(x)$  for all  $x \in Z$ . Since  $Z \subseteq X$ , we can find an extension (via the Hahn-Banach Theorem)  $\tilde{f}$  that is also a linear functional from

$Z$  to  $X$  satisfying  $\tilde{f}(x) \leq p(x)$  for all  $x \in X$ . All we need to show now is  $-p(-x) \leq \tilde{f}(x)$ . Since the bound in the previous statement holds for all  $x \in X$ , we have

$$-\tilde{f}(x) = \tilde{f}(-x) \leq p(-x).$$

Multiplying through by a negative, we now have

$$\tilde{f}(x) \geq -p(-x)$$

which completes our proof. ■

## 2 Week 7: Reflexive Spaces

**Theorem** (Dual Space (2.10-4)). The dual space  $X'$  of a normed space  $X$  is a Banach Space (whether or not  $X$  is).

**Proposition.** The dual space of  $\mathbb{R}^n$  is  $\mathbb{R}^n$ .

**Theorem** (3.8-1). Every bounded linear functional  $f$  on a Hilbert space  $H$  can be represented in terms of the inner product, namely,

$$f(x) = \langle x, z \rangle$$

where  $z$  depends on  $f$ , is uniquely determined by  $f$  and has norm

$$\|z\| = \|f\|.$$

Recall the definition of Algebraic Reflexivity of a vector space  $X$  that was introduced in section 2.8.

**Definition** (Algebraic Reflexivity). A vector space  $X$  is said to be **algebraic reflexive** if the canonical mapping  $C : X \rightarrow X^{**}$  is surjective.

- Here we see that  $X^{**}$  is the second algebraic dual space of  $X$  and the mapping  $C$  is defined by  $x \mapsto g_x$  where

$$g_x(f) = f(x). \quad ; \quad ((1) \ (f \in X^*))$$

that is, for any  $x \in X$  the image is the linear functional  $g_x$  defined by the above.

- If  $X$  is finite-dimensional, then  $X$  is algebraic reflexive.

Consider a normed space  $X$  and its dual space  $X'$  (the space of all bounded linear functionals). Observe that  $(X')'$  is the **second dual space** of  $X$  (or **bidual space** of  $X$ ). We define a similar mapping to the above statement, by setting

$$g_x(f) = f(x) \quad . \quad ((2), \ (f \in X'))$$

The only difference here is that  $f$  is now bounded and so  $g_x$  is bounded.

**Lemma** (Norm of  $g_x$ ). For every fixed  $x$  in a normed space  $X$ , the functional  $g_x$  defined above is a bounded linear functional on  $X'$ , so that  $g_x \in X''$ , and has the norm

$$\|g_x\| = \|x\|. \quad (3)$$

**Proof.** It is easy to see that  $g_x$  is linear, and (3) follows from (2) and corollary 4.3-4; that is,

$$\|g_x\| = \sup_{\substack{f \in X' \\ f \neq 0}} \frac{|g_x(f)|}{\|f\|} = \sup_{\substack{f \in X' \\ f \neq 0}} \frac{|f(x)|}{\|f\|} = \|x\|. \quad (4)$$

■

- This lemma tells us that for every  $x \in X$ , there corresponds a unique bounded linear functional  $g_x \in X''$  given by (2) which defines the **canonical mapping** of  $X$  into  $X''$ :

$$\begin{aligned} C : X &\longrightarrow X'' \\ x &\longmapsto g_x \end{aligned} \tag{5}$$

- This map is linear and injective and preserves the norm; that is, in terms of mapping the domain to its range, it will be an isomorphism of normed spaces.

**Lemma (Canonical Mapping).** The canonical mapping  $C$  given by (5) is an isomorphism of the normed space  $X$  onto the normed space  $\mathcal{R}(C)$ , the range of  $C$ .

**Proof.** We can easily see the linearity of  $C$ . Indeed, we have

$$g_{\alpha x + \beta y}(f) = f(\alpha x + \beta y) = \alpha f(x) + \beta f(y) = \alpha g_x(f) + \beta g_y(f).$$

In particular, we have  $g_x - g_y = g_{x-y}$ . Thus, we obtain

$$\|g_x - g_y\| = \|g_{x-y}\| = \|x - y\|.$$

This tells us that  $C$  is an isometry and so it must preserve the norm. Because isometries are injective,  $C$  is also injective. This can be directly seen from the above formula. Indeed, for any  $x \neq y$ , we have  $g_x \neq g_y$  by axiom (N2) in section 2.2. Since we are considering maps from  $X$  to the range of  $C$ , it follows that  $C$  is also surjective. Thus,  $C$  must be bijective. ■

**Definition (Embeddable).** We say that a set  $X$  is **embeddable** in a normed space  $Z$  if  $X$  is isomorphic with a subspace of  $Z$ .

- From our Canonical Mapping Lemma, we can see that  $X$  is embeddable in  $X''$  since the range of  $C$  is a subspace of  $X''$  and it is isomorphic to  $X$ .
- The mapping  $C$  is called the **canonical embedding** in this case.
- In general, the range of  $C$  will be a proper subspace of  $X''$ .
- If the range of  $C$  happens to be a surjection, then we will call it a different name which will be defined below.

**Definition (Reflexivity).** A normed space  $X$  is said to be **reflexive** if

$$\mathcal{R}(C) = X''$$

where  $C : X \rightarrow X''$  is the canonical mapping given by (5) and (2).

- If  $X$  is reflexive, it is isomorphic (and hence isometric) with  $X''$  by Lemma 4.6-2.
- However, the converse does not hold.
- Completeness does not necessarily imply reflexivity.
- However, the converse does hold which will be outlined below.

**Theorem (Completeness).** If a normed space  $X$  is reflexive, it is complete (hence a Banach space).

**Proof.** Since  $X''$  is the dual space of  $X'$ , it is complete by Theorem 2.10-4. Reflexivity of  $X$  means that  $\mathcal{R}(C) = X''$ . Completeness of  $X$  now follows from that of  $X''$  by Lemma 4.6-2. ■

- An example of a reflexive space is  $\mathbb{R}^n$  which follows from example 2.10-5.
- For finite dimensions, every linear functional on a general normed space  $X$  is bounded and so  $X' = X^*$  which implies further that

**Theorem (Finite Dimension).** Every finite-dimensional normed space is reflexive.

- $l^p$  with  $1 < p < +\infty$  is reflexive.
- $L^p[a, b]$  with  $1 < p < +\infty$  is reflexive.
- Non-examples are  $C[a, b]$ ,  $l^1$ ,  $L^1[a, b]$ ,  $l^\infty$  and the subspaces  $c$  and  $c_0$  of  $l^\infty$  where  $c$  is the space of all convergent sequence of scalars and  $c_0$  is the space of all sequences of scalars converging to zero.

**Theorem (Hilbert Space).** Every Hilbert Space  $H$  is reflexive.

**Proof.** Our goal is to show that the canonical mapping  $C : H \rightarrow H''$  is surjective by showing that for every  $g \in H''$ , there exists an  $x \in H$  such that  $g = Cx$ . Define  $A : H' \rightarrow H$  by  $Af = z$ , where  $z$  is given by the Riesz Representation  $f(x) = \langle x, z \rangle$ . From this, we see that  $A$  is bijective and isometric. Note that  $A$  is conjugate linear, as we see from (16) from section 4.5. Now,  $H'$  is complete by 2.10-4 and a Hilbert space with inner product defined by

$$\langle f_1, f_2 \rangle_1 = \langle Af_2, Af_1 \rangle.$$

It can be easily shown that  $\langle \cdot, \cdot \rangle_1$  is an inner product on  $H$ .

Let  $g \in H''$  be arbitrary. Let its Riesz Representation be

$$g(f) = \langle f, f_0 \rangle_1 = \langle Af_0, Af \rangle.$$

We now remember that  $f(x) = \langle x, z \rangle$  where  $z = Af$ . Writing  $Af_0 = x$ , we have

$$\langle Af_0, Af \rangle = \langle x, z \rangle = f(x).$$

Hence, we have  $g(f) = f(x)$  and so  $g = Cx$  by definition of  $C$ . Hence,  $C$  is surjective and so  $H$  must be reflexive. ■

- Separability of the dual of a normed space implies separability of  $X$  (note the converse is not generally true).
- If a normed space  $X$  is reflexive, then  $X''$  is isomorphic with  $X$  by 4.6-2, and so  $X$  implies separability of  $X''$ . Thus, the dual space  $X'$  must also be separable.
- A separable normed space  $X$  with a nonseparable dual space  $X'$  cannot be reflexive.

**Example** ( $l^1$  is not reflexive). Note that  $l^1$  is separable by 1.3-10, but the dual of  $l^1$  which is  $l^\infty$  is not via 2.10-6 and 1.3-9.

**Lemma (Existence of a Functional).** Let  $Y$  be a proper closed subspace of a normed space  $X$ . Let  $x_0 \in X \setminus Y$  be arbitrary and

$$\delta = \inf_{\tilde{y} \in Y} \|\tilde{y} - x_0\| \quad (6)$$

the distance from  $x_0$  to  $Y$ . Then there exists an  $\tilde{f} \in X'$  such that

$$\|\tilde{f}\| = 1 \quad \tilde{f}(y) = 0 \text{ for all } y \in Y, \quad \tilde{f}(x_0) = \delta. \quad (7)$$

**Proof.** Consider the subspace  $Z \subset X$  spanned by  $Y$  and  $x_0$  and define a bounded linear function  $f$  on  $Z$  by

$$f(z) = f(y + \alpha x_0) = \alpha \delta. \quad (y \in Y)$$

where  $z \in Z = \text{span}(Y \cup \{x_0\})$  has a unique representation

$$z = y + \alpha x_0 \quad y \in Y. \quad (8)$$

It is relatively easy to see that  $f$  is linear. Furthermore, because  $Y$  is a closed set and  $\delta > 0$ , we have  $f \neq 0$ . We may consider a few cases for  $\alpha$ . If  $\alpha = 0$ , then

$$f(y) = f(y + 0x_0) = 0 \cdot \delta = 0.$$

If  $\alpha = 1$  and  $y = 0$ , then we have  $f(x_0) = \delta$ . Now we show that  $f$  is bounded. If  $\alpha = 0$ , then  $f(z) = 0$ . If  $\alpha \neq 0$ , then since  $Y$  is a subspace of  $X$ , we know that  $-(1/\alpha)y \in Y$ . Using (6), we obtain the following inequality

$$\begin{aligned} |f(z)| &= |\alpha|\delta = |\alpha| \inf_{\tilde{y} \in Y} \|\tilde{y} - x_0\| \\ &\leq |\alpha| \left\| -\frac{1}{\alpha}y - x_0 \right\| \\ &= \|y + \alpha x_0\|, \end{aligned}$$

that is,  $|f(z)| \leq \|z\|$ . Thus, we see that  $f$  is bounded and so

$$\frac{|f(z)|}{\|z\|} \leq 1 \implies \|f\| \leq 1.$$

Next we show that  $\|f\| \geq 1$ . Using the definition of the infimum,  $Y$  contains a sequence  $(y_n)$  such that  $\|y_n - x_0\| \rightarrow \delta$ . Let  $z_n = y_n - x_0$ . Then we have  $f(z_n) = -\delta$  by (8) with  $\alpha = -1$ . Also

$$\|f\| = \sup_{\substack{z \in Z \\ z \neq 0}} \frac{|f(z)|}{\|z\|} \geq \frac{|f(z_n)|}{\|z_n\|} = \frac{\delta}{\|z_n\|} \xrightarrow{n \rightarrow \infty} \frac{\delta}{\delta} = 1.$$

Hence,  $\|f\| \geq 1$  and so we conclude that  $\|f\| = 1$ . Using the Hahn-Banach theorem for normed spaces we can extend  $f$  to  $X$  while preserving its norm. ■

**Theorem (Separability).** If the dual space  $X'$  of a normed space  $X$  is separable, then  $X$  itself is separable.

**Proof.** We assume that  $X'$  is separable. Then the unit sphere

$$U' = \{f : \|f\| = 1\} \subseteq X'$$

also contains a countable dense subset, say,  $(f_n)$ . Since  $f_n \in U'$ , we have

$$\|f_n\| = \sup_{\|x\|=1} \|f_n(x)\| = 1.$$

By the definition of the supremum, we can find points  $x_n \in X$  of norm 1 such that

$$\|f_n(x_n)\| \geq \frac{1}{2}.$$

Let  $Y$  be the closure of  $\text{span}(\{x_n\})$ . Then  $Y$  is separable because  $Y$  has a countable dense subset, namely, the set of all linear combinations of  $x_n$ 's with coefficients whose real and imaginary parts are rational.

Our goal is to show that  $Y = X$ . Suppose  $Y \neq X$ . Then, since  $Y$  is closed, by lemma 4.6-7 there exists an  $\tilde{f} \in X'$  with  $\|\tilde{f}\| = 1$  and  $\tilde{f}(y) = 0$  for all  $y \in Y$ . Since  $x_n \in Y$ , we have  $\tilde{f}(x_n) = 0$  and for all  $n$ ,

$$\begin{aligned} \frac{1}{2} &\leq |f_n(x_n)| = |f_n(x_n) - \tilde{f}(x_n)| \\ &= |(f_n - \tilde{f})(x_n)| \\ &\leq \|f_n - \tilde{f}\| \|x_n\| \end{aligned}$$

where  $\|x_n\| = 1$ . Hence,  $\|f_n - \tilde{f}\| \geq \frac{1}{2}$ , but this contradicts the assumption that  $(f_n)$  is dense in  $U'$  because  $\tilde{f}$  is itself in  $U'$  where  $\|\tilde{f}\| = 1$ . ■